

18 Operational amplifier (OPAMP)

18.1 OPAMP introduction

STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices embed 4 operational amplifiers OPAMP1, OPAMP2, OPAMP3 and OPAMP4 and STM32F303x6/8 and STM32F328x8 devices embed 1 operational amplifier OPAMP2. They can either be used as standalone amplifiers or as follower / programmable gain amplifiers.

The operational amplifier output is internally connected to an ADC channel for measurement purposes.

18.2 OPAMP main features

- Rail-to-rail input voltage range
- Low offset voltage
- Capability of being configured as a standalone operational amplifier or as a programmable gain amplifier (PGA)
- Access to all terminals
- Input multiplexer on inverting and non-inverting input
- Input multiplexer can be triggered by a timer and synchronized with a PWM signal.

18.3 OPAMP functional description

18.3.1 General description

On every OPAMP, there is one 4:1 multiplexer on the non-inverting input and one 2:1 multiplexer on the inverting input.

The inverting and non inverting inputs selection is made using the VM_SEL and VP_SEL bits respectively in the OPAMPx_CSR register.

The I/Os used as OPAMP input/outputs must be configured in analog mode in the GPIOs registers.

The connections with dedicated I/O are summarized in the table below and in [Figure 127](#), [Figure 128](#) and [Figure 129](#).

Table 110. Connections with dedicated I/O on STM32F303xB/C/D/E, STM32F358xC and STM32F398xE

OPAMP1 inverting input	OPAMP1 non inverting input	OPAMP2 inverting input	OPAMP2 non inverting input	OPAMP3 inverting input	OPAMP3 non inverting input	OPAMP4 inverting input	OPAMP4 non inverting input
PA3 (VM1)	PA1 (VP0)	PA5 (VM1)	PA7 (VP0)	PB2 (VM1)	PB0 (VP0)	PB10 (VM0)	PB13 (VP0)
PC5 (VM0)	PA7 (VP1)	PC5 (VM0)	PD14 (VP1)	PB10 (VM0)	PB13 (VP1)	PD8 (VM1)	PD11 (VP1)
-	PA3 (VP2)	-	PB0 (VP2)	-	PA1 (VP2)	-	PA4 (VP2)
-	PA5 (VP3)	-	PB14 (VP3)	-	PA5 (VP3)	-	PB11 (VP3)

Table 111. Connections with dedicated I/O on STM32F303x6/8 and STM32F328x8

OPAMP2 inverting input	OPAMP2 non inverting input
PA5 (VM1)	PA7 (VP0)
PC5 (VM0)	PD14 (VP1)
-	PB0 (VP2)

18.3.2 Clock

The OPAMP clock provided by the clock controller is synchronized with the PCLK2 (APB2 clock). There is no clock enable control bit provided in the RCC controller. To use a clock source for the OPAMP, the SYSCFG clock enable control bit must be set in the RCC controller.

18.3.3 Operational amplifiers and comparators interconnections

Internal connections between the operational amplifiers and the comparators are useful in motor control applications. These connections are summarized in the following figures.

Figure 127. STM32F303xB/C/D/E, STM32F358xC and STM32F398xE Comparators and operational amplifiers interconnections (part 1)

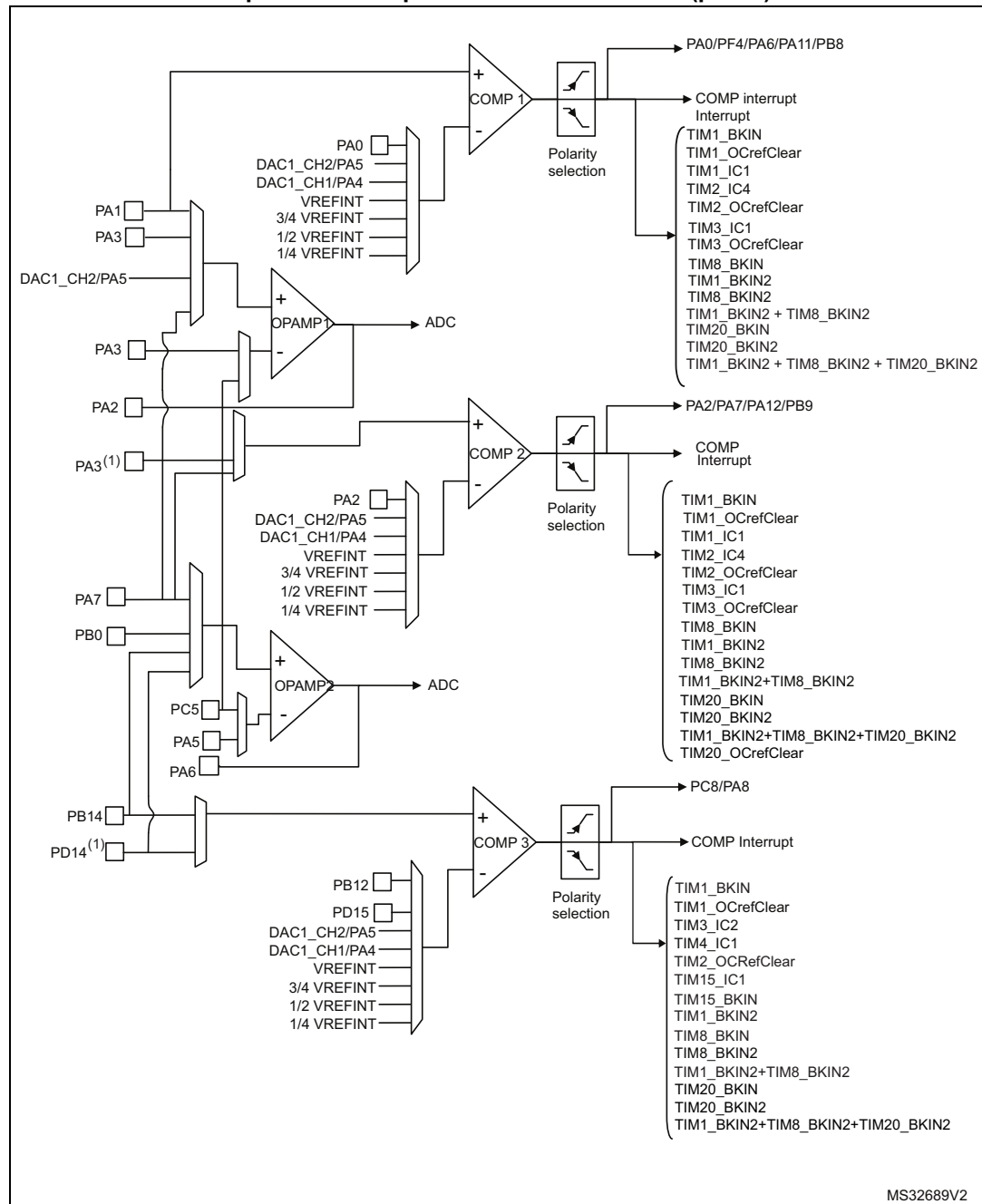
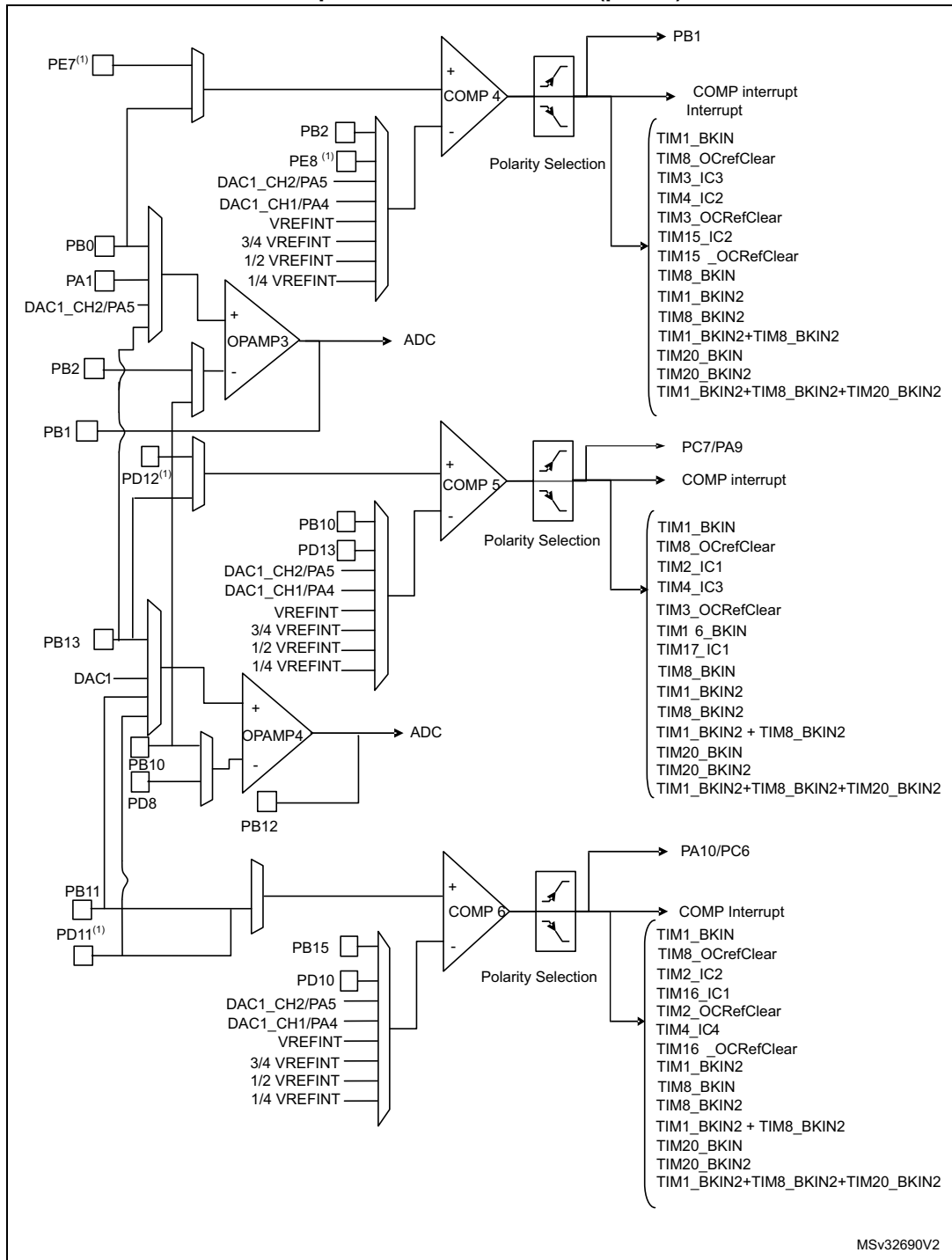
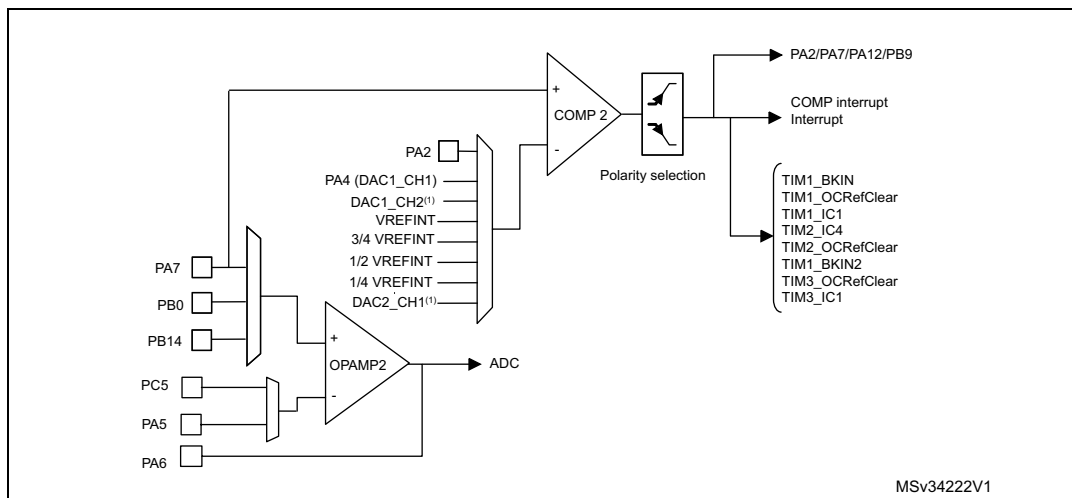


Figure 128. STM32F303xB/C/D/E and STM32F358xC comparators and operational amplifiers interconnections (part 2)



1. Only in STM32F303xB/C and STM32F358 devices.

Figure 129. STM32F303x6/8 and STM32F328x8 comparator and operational amplifier connections

1. DAC1_CH2 and DAC2_CH1 outputs are connected directly, thus PA5 and PA6 are not available as inputs as COMP2_INM. They can be used as GPIOs.

18.3.4 Using the OPAMP outputs as ADC inputs

In order to use OPAMP outputs as ADC inputs, the operational amplifiers must be enabled and the ADC must use the OPAMP output channel number:

- For OPAMP1, ADC1 channel 3 is used (only in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE).
- For OPAMP2, ADC2 channel 3 is used.
- For OPAMP3, ADC3 channel 1 is used (only in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE).
- For OPAMP4, ADC4 channel 3 is used (only in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE).

18.3.5 Calibration

The OPAMP interface continuously sends trimmed offset values to the 4 operational amplifiers. At startup, these values are initialized with the preset 'factory' trimming value.

Furthermore each operational amplifier offset can be trimmed by the user.

The user can switch from the 'factory' values to the 'user' trimmed values using the USER_TRIM bit in the OPAMP control register. This bit is reset at startup ('factory' values are sent to the operational amplifiers).

The rail-to-rail input stage of the OPAMP is composed of two differential pairs:

- One pair composed of NMOS transistors
- One pair composed of PMOS transistors.

As these two pairs are independent, the trimming procedure calibrates each one separately. The TRIMOFFSETN bits calibrate the NMOS differential pair offset and the TRIMOFFSETP bits calibrate the PMOS differential pair offset.

To calibrate the NMOS differential pair, the following conditions must be met: CALON=1 and CALSEL=11. In this case, an internal high voltage reference ($0.9 \times V_{DDA}$) is generated and

applied on the inverting and non inverting OPAMP inputs connected together. The voltage applied to both inputs of the OPAMP can be measured (the OPAMP reference voltage can be output through the TSTREF bit and connected internally to an ADC channel; refer to [Section 15: Analog-to-digital converters \(ADC\) on page 310](#)). The software should increment the TRIMOFFSETN bits in the OPAMP control register from 0x00 to the first value that causes the OUTCAL bit to change from 1 to 0 in the OPAMP register. If the OUTCAL bit is reset, the offset is calibrated correctly and the corresponding trimming value must be stored.

The calibration of the PMOS differential pair is performed in the same way, with two differences: the TRIMOFFSETP bits-fields are used and the CALSEL bits must be programmed to '01' (an internal low voltage reference ($0.1 \times V_{DDA}$) is generated and applied on the inverting and non inverting OPAMP inputs connected together).

Note: During calibration mode, to get the correct OUTCAL value, please make sure the OFFTRIMmax delay (specified in the datasheet electrical characteristics section) has elapsed between the write of a trimming value (TRIMOFFSETP or TRIMOFFSETN) and the read of the OUTCAL value,

To calibrate the NMOS differential pair, use the following software procedure:

1. Enable OPAMP by setting the OPAMPxEN bit
2. Enable the user offset trimming by setting the USERTRIM bit
3. Connect VM and VP to the internal reference voltage by setting the CALON bit
4. Set CALSEL to 11 (OPAMP internal reference $= 0.9 \times V_{DDA}$)
5. In a loop, increment the TRIMOFFSETN value. To exit from the loop, the OUTCAL bit must be reset. In this case, the TRIMOFFSETN value must be stored.

The same software procedure must be applied for PMOS differential pair calibration with CALSEL = 01 (OPAMP internal reference $= 0.1 \times V_{DDA}$).

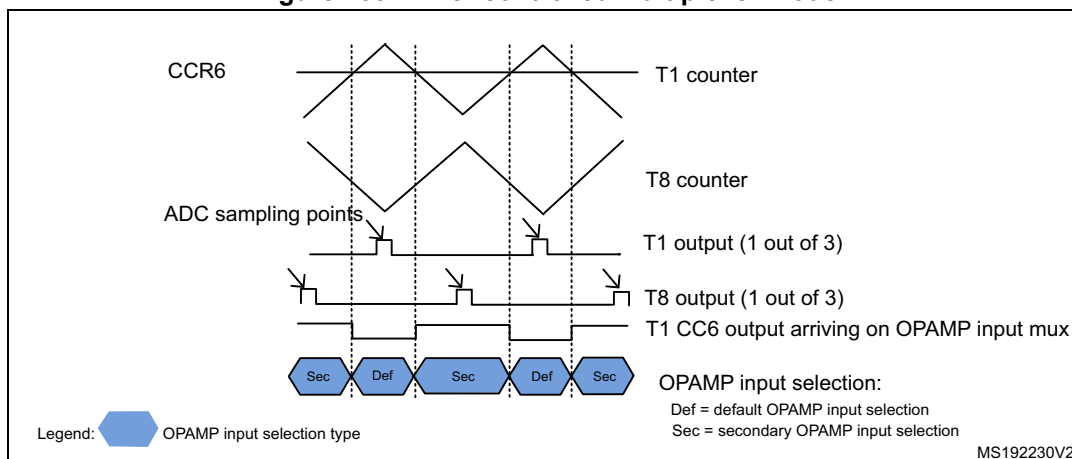
18.3.6 Timer controlled Multiplexer mode

The selection of the OPAMP inverting and non inverting inputs can be done automatically. In this case, the switch from one input to another is done automatically. This automatic switch is triggered by the TIM1 CC6 output arriving on the OPAMP input multiplexers.

This is useful for dual motor control with a need to measure the currents on the 3 phases instantaneously on a first motor and then on the second motor.

The automatic switch is enabled by setting the TCM_EN bit in the OPAMP control register. The inverting and non inverting inputs selection is performed using the VPS_SEL and VMS_SEL bit fields in the OPAMP control register. If the TCM_EN bit is cleared, the selection is done using the VP_SEL and VM_SEL bit fields in the OPAMP control register.

Figure 130. Timer controlled Multiplexer mode



18.3.7 OPAMP modes

The operational amplifier inputs and outputs are all accessible on terminals. The amplifiers can be used in multiple configuration environments:

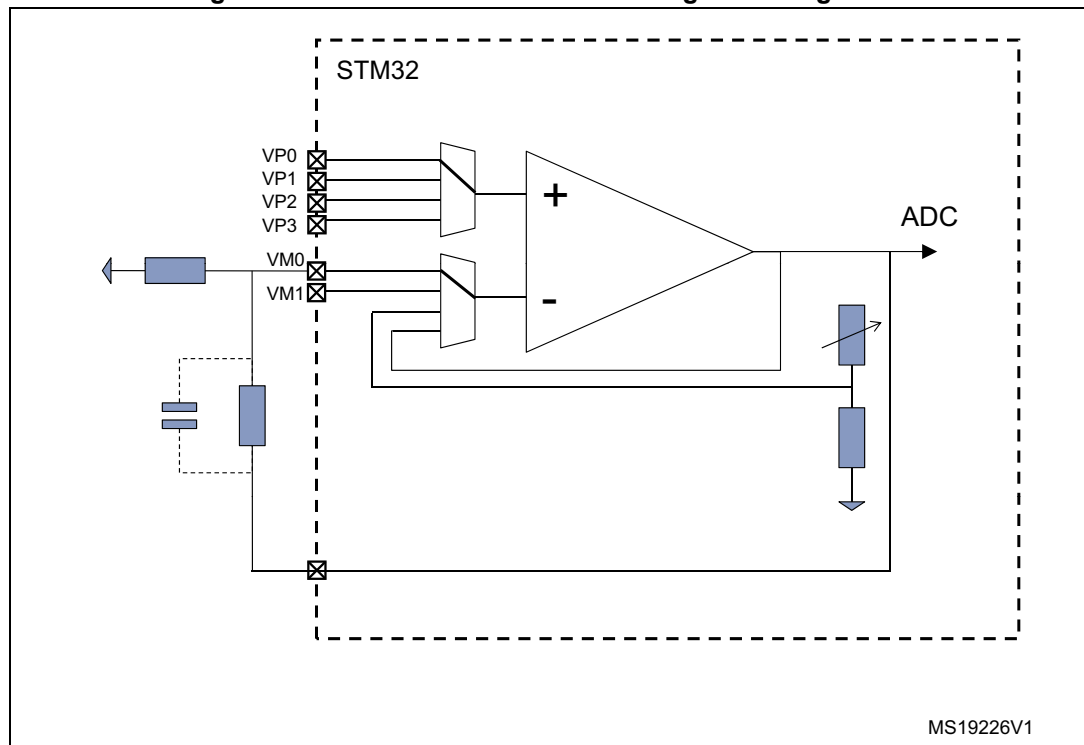
- Standalone mode (external gain setting mode)
- Follower configuration mode
- PGA modes

Important note: the amplifier output pin is directly connected to the output pad to minimize the output impedance. It cannot be used as a general purpose I/O, even if the amplifier is configured as a PGA and only connected to the ADC channel.

Note: The impedance of the signal must be maintained below a level which avoids the input leakage to create significant artefacts (due to a resistive drop in the source). Please refer to the electrical characteristics section in the datasheet for further details.

Standalone mode (external gain setting mode)

The external gain setting mode gives full flexibility to choose the amplifier configuration and feedback networks. This mode is enabled by writing the VM_SEL bits in the OPAMPx_CR register to 00 or 01, to connect the inverting inputs to one of the two possible I/Os.

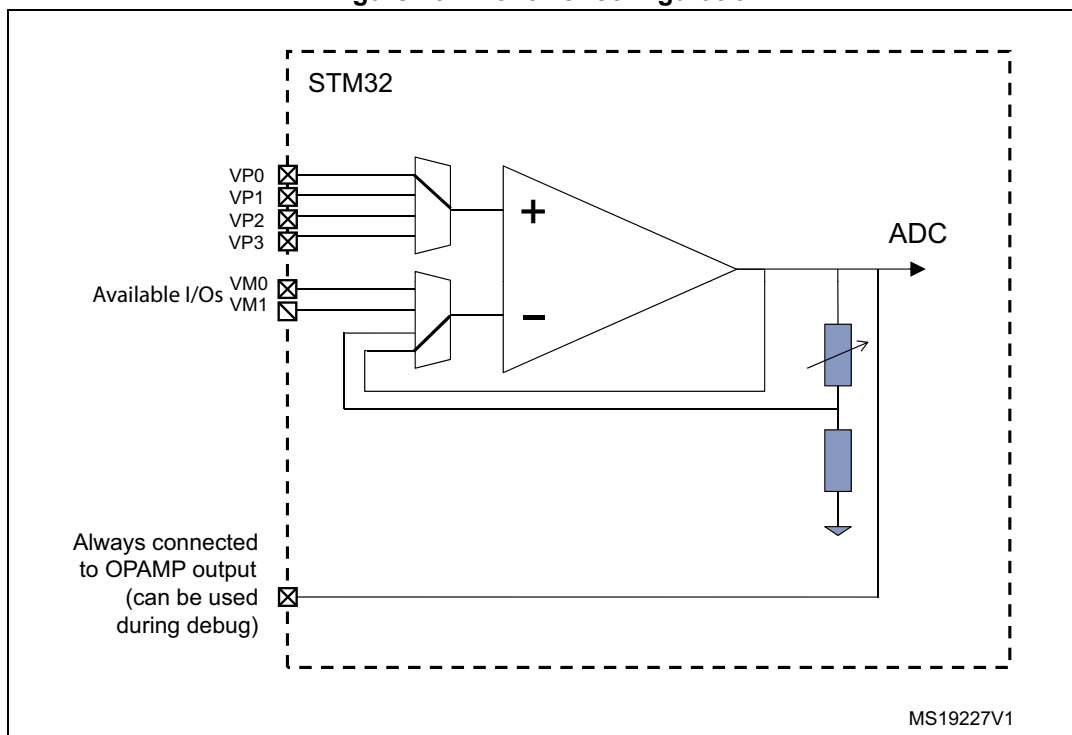
Figure 131. Standalone mode: external gain setting mode

1. This figure gives an example in an inverting configuration. Any other option is possible, including comparator mode.

Follower configuration mode

The amplifier can be configured as a follower, by setting the VM_SEL bits to 11 in the OPAMPx_CR register. This allows you for instance to buffer signals with a relatively high impedance. In this case, the inverting inputs are free and the corresponding ports can be used as regular I/Os.

Figure 132. Follower configuration



1. This figure gives an example in an inverting configuration. Any other option is possible, including comparator mode.

Programmable Gain Amplifier mode

The Programmable Gain Amplifier (PGA) mode is enabled by writing the VM_SEL bits to 10 in the OPAMPx_CR register. The gain is set using the PGA_GAIN bits which must be set to 0x00..0x11 for gains ranging from 2 to 16.

In this case, the inverting inputs are internally connected to the central point of a built-in gain setting resistive network. [Figure 133: PGA mode, internal gain setting \(x2/x4/x8/x16\), inverting input not used](#) shows the internal connection in this mode.

An alternative option in PGA mode allows you to route the central point of the resistive network on one of the I/Os connected to the non-inverting input. This is enabled using the PGA_GAIN bits in OPAMPx_CR register:

- 10xx values are setting the gain and connect the central point to one of the two available inputs
- 11xx values are setting the gain and connect the central point to the second available input

This feature can be used for instance to add a low-pass filter to PGA, as shown in [Figure 134: PGA mode, internal gain setting \(x2/x4/x8/x16\), inverting input used for filtering](#). Please note that the cut-off frequency is changed if the gain is modified (refer to the electrical characteristics section of the datasheet for details on resistive network elements).

Figure 133. PGA mode, internal gain setting (x2/x4/x8/x16), inverting input not used

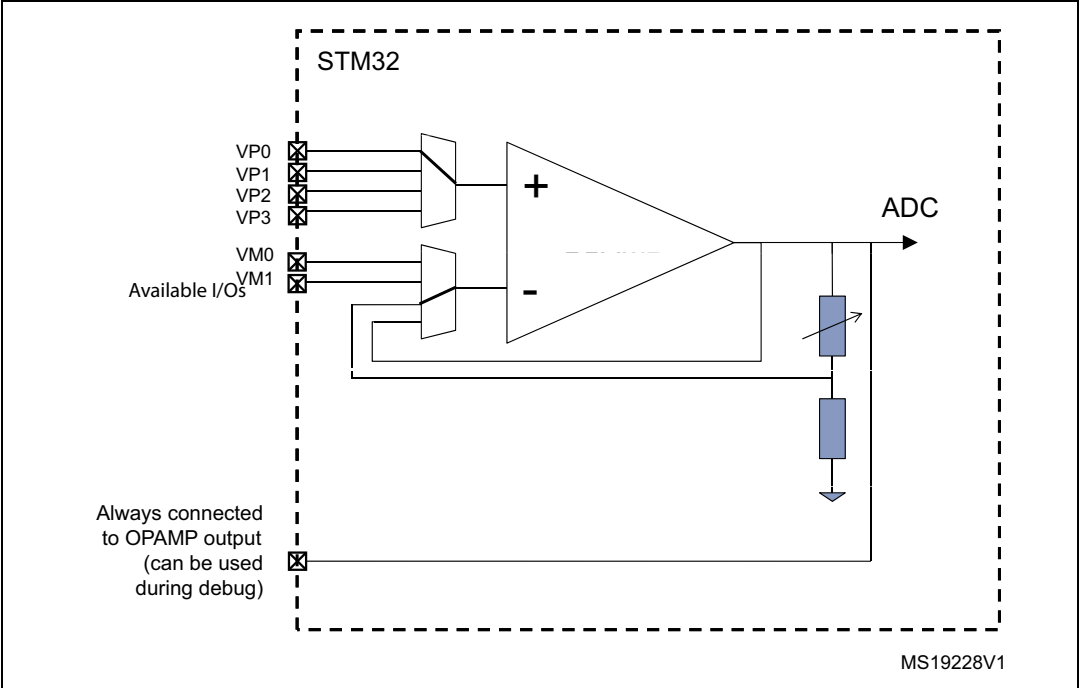
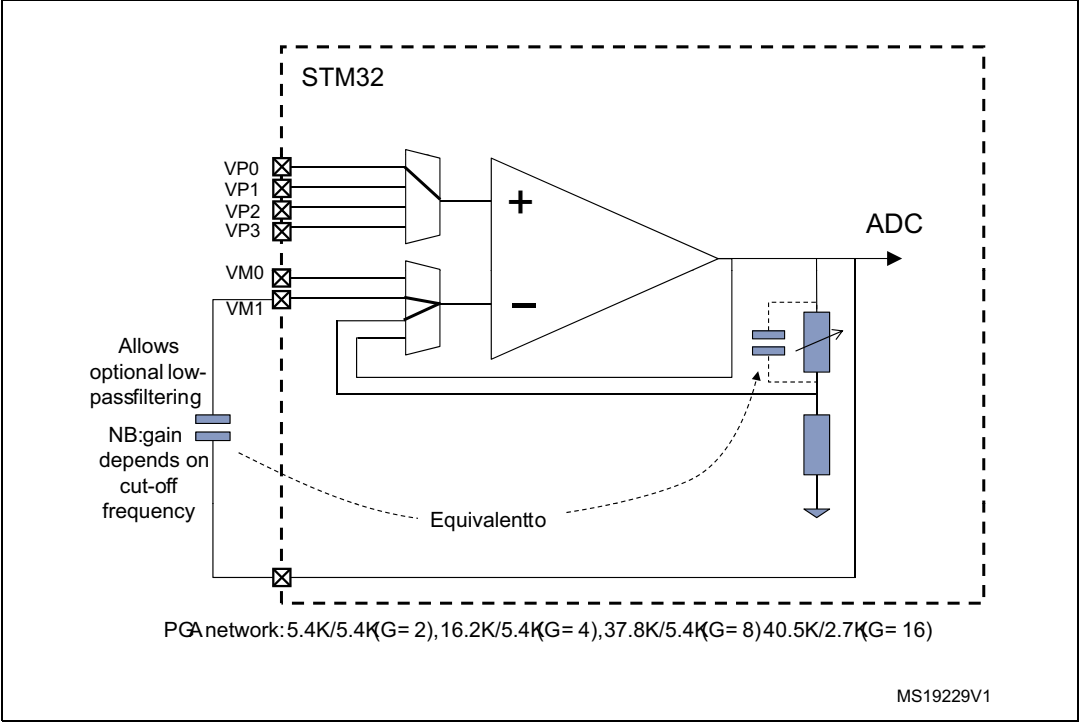


Figure 134. PGA mode, internal gain setting (x2/x4/x8/x16), inverting input used for filtering



18.4 OPAMP registers

18.4.1 OPAMP1 control register (OPAMP1_CSR)

Note: This register is only available in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices.

Address offset : 0x38

Reset value: 0xFFFF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCK	OUTCAL	TSTREF	TRIMOFFSETN					TRIMOFFSETP					USER_TRIM	PGA_GAIN	
rw	r	rw	rw					rw					rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PGA_GAIN		CALSEL		CALON	VPS_SEL		VMS_SEL	TCMEN	VM_SEL		Res.	VP_SEL		FORCE_VP	OPAMP1EN
rw		rw		rw	rw		rw	rw	rw			rw		rw	rw

Bit 31 **LOCK:** OPAMP 1 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

This bit is used to configure the OPAMP1_CSR register as read-only.

0: OPAMP1_CSR is read-write.

1: OPAMP1_CSR is read-only.

Bit 30 **OUTCAL:**

OPAMP output status flag, when the OPAMP is used as comparator during calibration.

0: Non-inverting < inverting

1: Non-inverting > inverting.

Bit 29 **TSTREF:**

This bit is set and cleared by software. It is used to output the internal reference voltage ($V_{REFOPAMP1}$).

0: $V_{REFOPAMP1}$ is output.

1: $V_{REFOPAMP1}$ is not output.

Bits 28:24 **TRIMOFFSETN:** Offset trimming value (NMOS)

Bits 23:19 **TRIMOFFSETP:** Offset trimming value (PMOS)

Bit 18 **USER_TRIM:** User trimming enable.

This bit is used to configure the OPAMP offset.

0: User trimming disabled.

1: User trimming enabled.

Bits 17:14 **PGA_GAIN**: Gain in PGA mode

- 0X00 = Non-inverting gain = 2
- 0X01 = Non-inverting gain = 4
- 0X10 = Non-inverting gain = 8
- 0X11 = Non-inverting gain = 16
- 1000 = Non-inverting gain = 2 - Internal feedback connected to VM0
- 1001 = Non-inverting gain = 4 - Internal feedback connected to VM0
- 1010 = Non-inverting gain = 8 - Internal feedback connected to VM0
- 1011 = Non-inverting gain = 16 - Internal feedback connected to VM0
- 1100 = Non-inverting gain = 2 - Internal feedback connected to VM1
- 1101 = Non-inverting gain = 4 - Internal feedback connected to VM1
- 1110 = Non-inverting gain = 8 - Internal feedback connected to VM1
- 1111 = Non-inverting gain = 16 - Internal feedback connected to VM1

Bits 13:12 **CALSEL**: Calibration selection

This bit is set and cleared by software. It is used to select the offset calibration bus used to generate the internal reference voltage when CALON = 1 or FORCE_VP = 1.

- 00 = $V_{REFOPAMP} = 3.3\% V_{DDA}$
- 01 = $V_{REFOPAMP} = 10\% V_{DDA}$
- 10 = $V_{REFOPAMP} = 50\% V_{DDA}$
- 11 = $V_{REFOPAMP} = 90\% V_{DDA}$

Bit 11 **CALON**: Calibration mode enable

This bit is set and cleared by software. It is used to enable the calibration mode connecting VM and VP to the OPAMP internal reference voltage.

- 0: Calibration mode disabled.
- 1: Calibration mode enabled.

Bits 10:9 **VPS_SEL**: OPAMP1 Non inverting input secondary selection.

These bits are set and cleared by software. They are used to select the OPAMP1 non inverting input when TCM_EN = 1.

- 00: PA7 used as OPAMP1 non inverting input
- 01: PA5 used as OPAMP1 non inverting input
- 10: PA3 used as OPAMP1 non inverting input
- 11: PA1 used as OPAMP1 non inverting input

Bit 8 **VMS_SEL**: OPAMP1 inverting input secondary selection

This bit is set and cleared by software. It is used to select the OPAMP1 inverting input when TCM_EN = 1.

- 0: PC5 (VM0) used as OPAMP1 inverting input
- 1: PA3 (VM1) used as OPAMP1 inverting input

Bit 7 **TCM_EN**: Timer controlled Mux mode enable

This bit is set and cleared by software. It is used to control automatically the switch between the default selection (VP_SEL and VM_SEL) and the secondary selection (VPS_SEL and VMS_SEL) of the inverting and non inverting inputs.

Bits 6:5 **VM_SEL**: OPAMP1 inverting input selection.

These bits are set and cleared by software. They are used to select the OPAMP1 inverting input.

- 00: PC5 (VM0) used as OPAMP1 inverting input
- 01: PA3 (VM1) used as OPAMP1 inverting input
- 10: Resistor feedback output (PGA mode)
- 11: follower mode

Bit 4 Reserved, must be kept at reset value.

Bits 3:2 **VP_SEL**: OPAMP1 Non inverting input selection.

These bits are set and cleared by software. They are used to select the OPAMP1 non inverting input.

00: PA7 used as OPAMP1 non inverting input

01: PA5 used as OPAMP1 non inverting input

10: PA3 used as OPAMP1 non inverting input

11: PA1 used as OPAMP1 non inverting input

Bit 1 **FORCE_VP**:

This bit forces a calibration reference voltage on non-inverting input and disables external connections.

0: Normal operating mode. Non-inverting input connected to inputs.

1: Calibration mode. Non-inverting input connected to calibration reference voltage.

Bit 0 **OPAMP1EN**: OPAMP1 enable.

This bit is set and cleared by software. It is used to enable the OPAMP1.

0: OPAMP1 is disabled.

1: OPAMP1 is enabled.

18.4.2 OPAMP2 control register (OPAMP2_CSR)

Address offset: 0x3C

Reset value: 0xFFFF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCK	OUTCAL	TSTREF	TRIMOFFSETN					TRIMOFFSETP					USER_TRIM	PGA_GAIN	
rw	r	rw	rw					rw					rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PGA_GAIN		CALSEL		CALON	VPS_SEL		VMS_SEL	TCM_EN	VM_SEL		Res.	VP_SEL		FORCE_VP	OPAMP2EN
rw		rw		rw	rw		rw	rw	rw			rw		rw	rw

Bit 31 **LOCK**: OPAMP 2 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

This bit is used to configure the OPAMP2_CSR register as read-only.

0: OPAMP2_CSR is read-write.

1: OPAMP2_CSR is read-only.

Bit 30 **OUTCAL**:

OPAMP output status flag, when the OPAMP is used as comparator during calibration.

0: Non-inverting < inverting

1: Non-inverting > inverting.

Bit 29 **TSTREF**:

This bit is set and cleared by software. It is used to output the internal reference voltage ($V_{REFOPAMP2}$).

0: $V_{REFOPAMP2}$ is output.

1: $V_{REFOPAMP2}$ is not output.

Bits 28:24 **TRIMOFFSETN**: Offset trimming value (NMOS)

Bits 23:19 **TRIMOFFSETP**: Offset trimming value (PMOS)

Bit 18 **USER_TRIM**: User trimming enable.

This bit is used to configure the OPAMP offset.

0: User trimming disabled.

1: User trimming enabled.

Bits 17:14 **PGA_GAIN**: gain in PGA mode

0X00 = Non-inverting gain = 2

0X01 = Non-inverting gain = 4

0X10 = Non-inverting gain = 8

0X11 = Non-inverting gain = 16

1000 = Non-inverting gain = 2 - Internal feedback connected to VM0

1001 = Non-inverting gain = 4 - Internal feedback connected to VM0

1010 = Non-inverting gain = 8 - Internal feedback connected to VM0

1011 = Non-inverting gain = 16 - Internal feedback connected to VM0

1100 = Non-inverting gain = 2 - Internal feedback connected to VM1

1101 = Non-inverting gain = 4 - Internal feedback connected to VM1

1110 = Non-inverting gain = 8 - Internal feedback connected to VM1

1111 = Non-inverting gain = 16 - Internal feedback connected to VM1

Bits 13:12 **CALSEL**: Calibration selection

This bit is set and cleared by software. It is used to select the offset calibration bus used to generate the internal reference voltage when CALON = 1 or FORCE_VP = 1.

00 = $V_{REFOPAMP} = 3.3\% V_{DDA}$

01 = $V_{REFOPAMP} = 10\% V_{DDA}$

10 = $V_{REFOPAMP} = 50\% V_{DDA}$

11 = $V_{REFOPAMP} = 90\% V_{DDA}$

Bit 11 **CALON**: Calibration mode enable

This bit is set and cleared by software. It is used to enable the calibration mode connecting VM and VP to the OPAMP internal reference voltage.

0: calibration mode disabled.

1: calibration mode enabled.

Bits 10:9 **VPS_SEL**: OPAMP2 Non inverting input secondary selection.

These bits are set and cleared by software. They are used to select the OPAMP2 non inverting input when TCM_EN = 1.

00: PD14 used as OPAMP2 non inverting input (STM32F303xB/C and STM32F358C devices only)

01: PB14 used as OPAMP2 non inverting input

10: PB0 used as OPAMP2 non inverting input

11: PA7 used as OPAMP2 non inverting input

Bit 8 **VMS_SEL**: OPAMP2 inverting input secondary selection

This bit is set and cleared by software. It is used to select the OPAMP2 inverting input when TCM_EN = 1.

0: PC5 (VM0) used as OPAMP2 inverting input

1: PA5 (VM1) used as OPAMP2 inverting input

Bit 7 **TCM_EN**: Timer controlled Mux mode enable.

This bit is set and cleared by software. It is used to control automatically the switch between the default selection (VP_SEL and VM_SEL) and the secondary selection (VPS_SEL and VMS_SEL) of the inverting and non inverting inputs.

Bits 6:5 **VM_SEL**: OPAMP2 inverting input selection.

These bits are set and cleared by software. They are used to select the OPAMP2 inverting input.

00: PC5 (VM0) used as OPAMP2 inverting input

01: PA5 (VM1) used as OPAMP2 inverting input

10: Resistor feedback output (PGA mode)

11: follower mode

Bit 4 Reserved, must be kept at reset value.

Bits 3:2 **VP_SEL**: OPAMP2 non inverting input selection.

These bits are set/reset by software. They are used to select the OPAMP2 non inverting input.

00: PD14 used as OPAMP2 non inverting input (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

01: PB14 used as OPAMP2 non inverting input

10: PB0 used as OPAMP2 non inverting input

11: PA7 used as OPAMP2 non inverting input

Bit 1 **FORCE_VP**:

This bit forces a calibration reference voltage on non-inverting input and disables external connections.

0: Normal operating mode. Non-inverting input connected to inputs.

1: Calibration mode. Non-inverting input connected to calibration reference voltage.

Bit 0 **OPAMP2EN**: OPAMP2 enable.

This bit is set and cleared by software. It is used to select the OPAMP2.

0: OPAMP2 is disabled.

1: OPAMP2 is enabled.

18.4.3 OPAMP3 control register (OPAMP3_CSR)

Note: This register is only available in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices.

Address offset: 0x40

Reset value: 0xFFFF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCK	OUT CAL	TSTR EF	TRIMOFFSETN					TRIMOFFSETP					USER_ TRIM	PGA_GAIN	
rw	r	rw	rw					rw					rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PGA_GAIN		CALSEL		CAL ON	VPS_SEL		VMS_ SEL	TCM_ EN	VM_SEL		Res.	VP_SEL		FORCE_ _VP	OPAMP 3EN
rw		rw		rw	rw		rw	rw	rw			rw		rw	rw

Bit 31 **LOCK:** OPAMP 3 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

This bit is used to configure the OPAMP3_CSR register as read-only.

0: OPAMP3_CSR is read-write.

1: OPAMP3_CSR is read-only.

Bit 30 **OUTCAL:**

OPAMP output status flag, when the OPAMP is used as comparator during calibration.

0: Non-inverting < inverting

1: Non-inverting > inverting.

Bit 29 **TSTREF:**

This bit is set and cleared by software. It is used to output the internal reference voltage ($V_{REFOPAMP3}$).

0: $V_{REFOPAMP3}$ is output.

1: $V_{REFOPAMP3}$ is not output.

Bits 28:24 **TRIMOFFSETN:** Offset trimming value (NMOS)

Bits 23:19 **TRIMOFFSETP:** Offset trimming value (PMOS)

Bit 18 **USER_TRIM:** User trimming enable.

This bit is used to configure the OPAMP offset.

0: User trimming disabled.

1: User trimming enabled.

Bits 17:14 **PGA_GAIN:** gain in PGA mode

0X00 = Non-inverting gain = 2

0X01 = Non-inverting gain = 4

0X10 = Non-inverting gain = 8

0X11 = Non-inverting gain = 16

1000 = Non-inverting gain = 2 - Internal feedback connected to VM0

1001 = Non-inverting gain = 4 - Internal feedback connected to VM0

1010 = Non-inverting gain = 8 - Internal feedback connected to VM0

1011 = Non-inverting gain = 16 - Internal feedback connected to VM0

1100 = Non-inverting gain = 2 - Internal feedback connected to VM1

1101 = Non-inverting gain = 4 - Internal feedback connected to VM1

1110 = Non-inverting gain = 8 - Internal feedback connected to VM1

1111 = Non-inverting gain = 16 - Internal feedback connected to VM1

Bits 13:12 **CALSEL:** Calibration selection

This bit is set and cleared by software. It is used to select the offset calibration bus used to generate the internal reference voltage when CALON = 1 or FORCE_VP = 1.

00 = $V_{REFOPAMP} = 3.3\% V_{DDA}$

01 = $V_{REFOPAMP} = 10\% V_{DDA}$

10 = $V_{REFOPAMP} = 50\% V_{DDA}$

11 = $V_{REFOPAMP} = 90\% V_{DDA}$

Bit 11 **CALON:** Calibration mode enable

This bit is set/cleared by software. It allows enabling the calibration mode connecting VM and VP to internal reference voltage.

0: calibration mode disabled.

1: calibration mode enabled.

Bits 10:9 **VPS_SEL**: OPAMP3 non inverting input secondary selection.

These bits are set/reset by software. They allow selecting the OPAMP3 non inverting input when TCM_EN = 1.

00: PB13 used as OPAMP3 non inverting input

01: PA5 used as OPAMP3 non inverting input

10: PA1 used as OPAMP3 non inverting input

11: PB0 used as OPAMP3 non inverting input

Bit 8 **VMS_SEL**: OPAMP3 inverting input secondary selection

This bit is set and cleared by software. It is used to select the OPAMP3 inverting input when TCM_EN = 1.

0: PB10 (VM0) used as OPAMP3 inverting input

1: PB2 (VM1) used as OPAMP3 inverting input

Bit 7 **TCM_EN**: Timer controlled multiplexer mode enable.

This bit is set and cleared by software. It is used to control automatically the switch between the default selection (VP_SEL and VM_SEL) and the secondary selection (VPS_SEL and VMS_SEL) of the inverting and non inverting inputs.

Bits 6:5 **VM_SEL**: OPAMP3 inverting input selection.

These bits are set/reset by software. They allow selecting the OPAMP3 inverting input.

00: PB10 (VM0) used as OPAMP3 inverting input

01: PB2 (VM1) used as OPAMP3 inverting input

10: Resistor feedback output (PGA mode)

11: follower mode

Bit 4 Reserved, must be kept at reset value.

Bits 3:2 **VP_SEL**: OPAMP3 Non inverting input selection.

These bits are set and cleared by software. They are used to select the OPAMP3 non inverting input.

00: PB13 used as OPAMP3 non inverting input

01: PA5 used as OPAMP3 non inverting input

10: PA1 used as OPAMP3 non inverting input

11: PB0 used as OPAMP3 non inverting input

Bit 1 **FORCE_VP**:

This bit forces a calibration reference voltage on non-inverting input and disables external connections.

0: Normal operating mode. Non-inverting input connected to inputs.

1: Calibration mode. Non-inverting input connected to calibration reference voltage.

Bit 0 **OPAMP3EN**: OPAMP3 enable.

This bit is set and cleared by software. It is used to enable the OPAMP3.

0: OPAMP3 is disabled.

1: OPAMP3 is enabled.

18.4.4 OPAMP4 control register (OPAMP4_CSR)

Note: This register is only available in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices.

Address offset: 0x44

Reset value: 0xFFFF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
LOCK	OUT CAL	TSTR EF	TRIMOFFSETN						TRIMOFFSETP						USER_ TRIM	PGA_GAIN
rw	r	rw	rw						rw						rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
PGA_GAIN		CALSEL		CAL ON	VPS_SEL		VMS_ SEL	TCM_ EN	VM_SEL		Res.	VP_SEL		FORCE_ VP	OPAMP 4EN	
rw		rw		rw	rw		rw	rw	rw			rw		rw	rw	

Bit 31 **LOCK:** OPAMP 4 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

This bit is used to configure the OPAMP4_CSR register as read-only.

0: OPAMP4_CSR is read-write.

1: OPAMP4_CSR is read-only.

Bit 30 **OUTCAL:**

OPAMP output status flag, when the OPAMP is used as comparator during calibration.

0: Non-inverting < inverting

1: Non-inverting > inverting.

Bit 29 **TSTREF:**

This bit is set and cleared by software. It is used to output the internal reference voltage ($V_{REFOPAMP4}$).

0: $V_{REFOPAMP4}$ is output.

1: $V_{REFOPAMP4}$ is not output.

Bits 28:24 **TRIMOFFSETN:** Offset trimming value (NMOS)

Bits 23:19 **TRIMOFFSETP:** Offset trimming value (PMOS)

Bit 18 **USER_TRIM:** User trimming enable.

This bit is used to configure the OPAMP offset.

0: User trimming disabled.

1: User trimming enabled.

Bits 17:14 **PGA_GAIN**: Gain in PGA mode

0X00 = Non-inverting gain = 2
 0X01 = Non-inverting gain = 4
 0X10 = Non-inverting gain = 8
 0X11 = Non-inverting gain = 16
 1000 = Non-inverting gain = 2 - Internal feedback connected to VM0
 1001 = Non-inverting gain = 4 - Internal feedback connected to VM0
 1010 = Non-inverting gain = 8 - Internal feedback connected to VM0
 1011 = Non-inverting gain = 16 - Internal feedback connected to VM0
 1100 = Non-inverting gain = 2 - Internal feedback connected to VM1
 1101 = Non-inverting gain = 4 - Internal feedback connected to VM1
 1110 = Non-inverting gain = 8 - Internal feedback connected to VM1
 1111 = Non-inverting gain = 16 - Internal feedback connected to VM1

Bits 13:12 **CALSEL**: Calibration selection

This bit is set and cleared by software. It is used to select the offset calibration bus used to generate the internal reference voltage when CALON = 1 or FORCE_VP= 1.

00 = $V_{REFOPAMP} = 3.3\% V_{DDA}$
 01 = $V_{REFOPAMP} = 10\% V_{DDA}$
 10 = $V_{REFOPAMP} = 50\% V_{DDA}$
 11 = $V_{REFOPAMP} = 90\% V_{DDA}$

Bit 11 **CALON**: Calibration mode enable

This bit is set and cleared by software. It is used to enable the calibration mode connecting VM and VP to the OPAMP internal reference voltage.

0: Calibration mode disabled.
 1: Calibration mode enabled.

Bits 10:9 **VPS_SEL**: OPAMP4 Non inverting input secondary selection.

These bits are set and cleared by software. They allow selecting the OPAMP4 non inverting input, when TCM_EN = 1.

00: PD11 used as OPAMP4 non inverting input
 01: PB11 used as OPAMP4 non inverting input
 10: PA4 used as OPAMP4 non inverting input
 11: PB13 used as OPAMP4 non inverting input

Bit 8 **VMS_SEL**: OPAMP4 inverting input secondary selection

This bit is set and cleared by software. It allows selecting the OPAMP4 inverting input, when TCM_EN = 1.

0: PB10 (VM0) used as OPAMP4 inverting input
 1: PD8 (VM1) used as OPAMP4 inverting input

Bit 7 **TCM_EN**: Timer controlled Mux mode enable

This bit is set and cleared by software. It is used to control automatically the switch between the default selection (VP_SEL and VM_SEL) and the secondary selection (VPS_SEL and VMS_SEL) of the inverting and non inverting inputs.

Bits 6:5 **VM_SEL**: OPAMP4 inverting input selection.

These bits are set/reset by software. They allow selecting the OPAMP4 inverting input.

00: PB10 (VM0) used as OPAMP4 inverting input
 01: PD8 (VM1) used as OPAMP4 inverting input
 10: Resistor feedback output (PGA mode)
 11: follower mode

Bit 4 Reserved, must be kept at reset value.

Bits 3:2 **VP_SEL**: OPAMP4 Non inverting input selection.

These bits are set and cleared by software. They allow selecting the OPAMP4 non inverting input.

00: PD11 used as OPAMP4 non inverting input

01: PB11 used as OPAMP4 non inverting input

10: PA4 used as OPAMP4 non inverting input

11: PB13 used as OPAMP4 non inverting input

Bit 1 **FORCE_VP**:

This description will be given in a future version of this document.

Bit 0 **OPAMP4EN**: OPAMP4 enable.

This bit is set and cleared by software. It allows enabling the OPAMP4.

0: OPAMP4 is disabled.

1: OPAMP4 is enabled.

18.4.5 OPAMP register map

The following table summarizes the OPAMP registers.

Table 112. OPAMP register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x38	OPAMP1_CSR	LOCK	OUTCAL	TSTREF			TRIMOFFSETN					TRIMOFFSETP			USER_TRIM		PGA_GAIN			CALSEL		CALON	VPS_SEL	VMS_SEL	TCM_EN	VM_SEL		Res		VP_SEL	FORCE_VP	OPAMP1EN	
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0
0x3C	OPAMP2_CSR	LOCK	OUTCAL	TSTREF			TRIMOFFSETN					TRIMOFFSETP			USER_TRIM		PGA_GAIN			CALSEL		CALON	VPS_SEL	VMS_SEL	TCM_EN	VM_SEL		Res		VP_SEL	FORCE_VP	OPAMP2EN	
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x40	OPAMP3_CSR	LOCK	OUTCAL	TSTREF			TRIMOFFSETN					TRIMOFFSETP			USER_TRIM		PGA_GAIN			CALSEL		CALON	VPS_SEL	VMS_SEL	TCM_EN	VM_SEL		Res		VP_SEL	FORCE_VP	OPAMP3EN	
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0
0x44	OPAMP4_CSR	LOCK	OUTCAL	TSTREF			TRIMOFFSETN					TRIMOFFSETP			USER_TRIM		PGA_GAIN			CALSEL		CALON	VPS_SEL	VMS_SEL	TCM_EN	VM_SEL		Res		VP_SEL	FORCE_VP	OPAMP4EN	
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.